

Lab Problem 4: R Project Implementation and Version Control Using GitHub

Problem Statement

You are required to study the project demonstrated in the prescribed video tutorial and implement the project using **R Programming** by following the concepts, procedures, and implementation steps explained in the video.

After completing the project, test the implementation for correctness and maintain the complete project using **Git for version control**. Create a repository on your **GitHub account**, upload/push the complete project, and appropriately document the implementation using a **README.md** file.

Reference Video

https://www.youtube.com/watch?v=iExh0qj2Ouo&list=PLpaxqU9O4beceS4RxsJESmFGUqpw-m2_P&index=4

Tasks to be Performed

1. Watch the prescribed video and understand the problem, dataset, R programming concepts, libraries, and implementation procedure demonstrated.
2. Set up the required R/RStudio environment and install/load the necessary R packages.
3. Implement the complete project demonstrated in the video using **R Programming**.
4. Execute and test the developed project and verify the correctness of the generated results.
5. Organize the R scripts, dataset files, generated outputs, and other supporting files systematically.
6. Create a **Git repository** for the project and maintain the project using appropriate Git operations.
7. Make meaningful commits at different stages of project development instead of uploading the complete project in a single commit.
8. Create a repository on your **GitHub account** and push the completed project.
9. Prepare a **README.md** file containing:
 - Project title and objective
 - Brief description of the problem
 - Dataset information
 - R packages/libraries used
 - Major operations performed
 - Instructions to execute the project
 - Important results/output
 - Screenshots, wherever appropriate
10. Submit the GitHub repository link for evaluation.

Deliverables

- Complete R project/source code

- Dataset used in the project or appropriate dataset reference
- Output/results generated by the project
- README.md file
- GitHub repository link
- Meaningful Git commit history
- Screenshot(s) demonstrating successful execution

AI Feedback Scores

Rating: 1 – Lowest | 4 – Highest

Parameter	Score
Real-life Application	4 / 4
Problem-solving Ability	4 / 4
Critical Reasoning	3 / 4
Creativity	3 / 4
Alignment with Learning / Course Outcomes	4 / 4

Estimated Time Required

1 Hour 45 Minutes – 2 Hours

Suggested Time Distribution

- **15–20 Minutes:** Study the prescribed video and understand the project requirements.
- **10–15 Minutes:** Configure the R/RStudio environment, dataset, and required packages.
- **45–55 Minutes:** Implement the project using R.
- **15–20 Minutes:** Execute, test, debug, and validate the results.
- **15–20 Minutes:** Create the Git repository, maintain commits, prepare README.md, and push the project to GitHub.

COs Linked with the Assignment

CO3 and CO6

CO	Course Outcome	Level of Alignment
CO3	Apply R programming to import/export data, perform statistical tests, and implement basic machine learning models.	Strong
CO6	Design and implement an end-to-end data science project using R, from data cleaning to model deployment.	Strong

Primary CO: CO6

LOs Linked with the Assignment

LO 3.1, LO 3.2, LO 6.1, LO 6.2

Suggested Learning Outcomes

- **LO 3.1:** Apply appropriate R programming techniques for importing, processing, and analyzing project data.
- **LO 3.2:** Implement the required analytical/statistical operations using appropriate R packages and functions.
- **LO 6.1:** Develop an end-to-end data science project by integrating data handling, analysis, visualization, and result generation.
- **LO 6.2:** Organize, document, version-control, and publish an R-based data science project using Git and GitHub.

Justification

This laboratory problem provides students with practical exposure to **project-based R programming and end-to-end data science implementation**. Students are required to study a prescribed technical demonstration, understand the project workflow, configure the required programming environment, implement the solution using R, test the generated results, and systematically publish the completed project through GitHub. The activity therefore moves beyond isolated R commands and introduces students to a structured project-development workflow.

The assignment has a high level of **real-life application** because modern data science projects require programmers to combine data processing and analytical programming with testing, documentation, reproducibility, and version control. The use of **Git and GitHub** further exposes students to professional practices commonly followed in academic, research, and industrial software and data science projects.

The activity strongly develops **problem-solving ability** because students must understand the requirements demonstrated in the tutorial, identify the required R packages and functions, implement the solution, handle possible programming or data-related errors, and validate the final results. Students are therefore required to integrate several practical skills rather than merely execute predefined individual R commands.

The assignment supports **critical reasoning** through interpretation of the project workflow, selection and application of appropriate R operations, debugging of implementation errors, verification of generated outputs, and evaluation of whether the developed solution meets the expected project requirements. Since the primary implementation is guided by a prescribed tutorial, the level of independent analytical decision-making is comparatively moderate; hence a score of **3/4** is appropriate.

The assignment provides a moderate level of **creativity** because students follow a prescribed project implementation. However, students can demonstrate creativity through project organization, code structure, documentation, presentation of results, README design, visualization of outputs, and improvements to the demonstrated implementation. An optional additional feature can be introduced if a higher level of creativity is desired.

The assignment has particularly strong alignment with **CO3** and **CO6**. **CO3** is addressed through the practical application of R programming for working with data and implementing the analytical operations required by the project. Depending upon the exact project demonstrated in the prescribed video, this may include data import/export, data processing, statistical operations, or basic analytical/modeling tasks.

The strongest alignment is with **CO6**, which requires students to **design and implement an end-to-end data science project using R, from data cleaning to model deployment**. The laboratory activity requires students to progress through multiple stages of a practical project, including understanding requirements, preparing the environment, handling data, implementing the solution, generating results, testing the implementation, documenting the project, and publishing the final work through GitHub. Therefore, **CO6 should be considered the primary Course Outcome mapped to this laboratory problem**.

The activity does not directly assess **CO1, CO2, CO4, or CO5** unless the project demonstrated in the prescribed video explicitly involves those concepts. Therefore, these COs should not be mapped merely for the purpose of increasing CO coverage. This ensures that the CO mapping remains academically valid and based on the actual student tasks performed.

AI Tool Used

ChatGPT

Prompt(s) Used

"Evaluate the given R Programming laboratory problem in which students study a prescribed project video, implement the demonstrated project using R, test the implementation, maintain the project using Git, and publish it on GitHub. Evaluate the problem using a 1–4 rubric for Real-life Application, Problem-solving Ability, Critical Reasoning, Creativity, and Alignment with Learning/Course Outcomes. Estimate the time required and map the assignment specifically against the following R Programming Course Outcomes: CO1 – fundamental R concepts and data structures; CO2 – data manipulation, visualization and statistical concepts; CO3 – application of R for data import/export, statistical tests and basic machine learning; CO4 – model evaluation and code optimization; CO5 – comparison and selection of machine learning algorithms; and CO6 – design and implementation of an end-to-end data science project using R. Identify only those COs genuinely addressed by the laboratory activity and provide appropriate Learning Outcomes and detailed justification for the mapping."